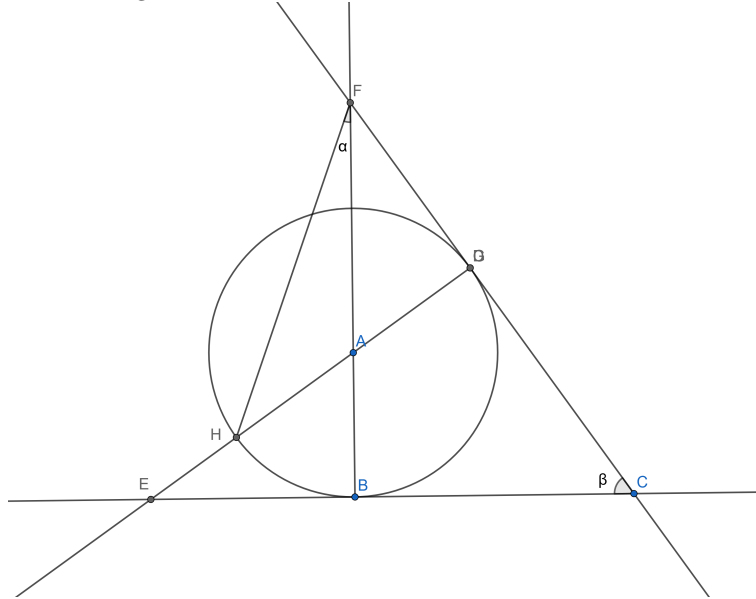


A geometry problem solved by differentiation (circle, tangent, differentiation (trigo))

In the figure, A is the centre of the circle. EC and FC are tangents to the circle at B and G respectively. BAF is a straight line.



Find the maximum of angle α and the corresponding angle β .

Solution

In $\triangle ABE$, $\angle BAE = \beta$.

In $\triangle FAG$, $\angle FAG = \beta$.

Let $FG = h$. Let $\angle FHA = \gamma$.

$$\tan \beta = \frac{h}{r}$$

$$\tan \gamma = \frac{h}{2r} = \frac{\tan \beta}{2}$$

$$\alpha + \gamma = \beta$$

$$\alpha = \beta - \gamma$$

$$\tan \alpha = \tan(\beta - \gamma)$$

$$= \frac{\tan \beta - \frac{\tan \beta}{2}}{1 + \tan \beta \times \frac{\tan \beta}{2}}$$

$$= \frac{\tan \beta}{2 + \tan^2 \beta}$$

Let $y = \frac{x}{2+x^2}$.

Differentiate:

$$\frac{dy}{dx} = \frac{(2+x^2) - 2x^2}{(2+x^2)^2}$$

$$= \frac{2-x^2}{(2+x^2)^2}$$

Set derivative to zero:

$$2 - x^2 = 0 \quad \Rightarrow \quad x = \pm\sqrt{2}$$

Second derivative test:

$$\begin{aligned}\frac{d^2y}{dx^2} &= \frac{(-2x)(2+x^2)^2 - (2-x^2)[2(2+x^2)(2x)]}{(2+x^2)^4} \\ &= \frac{(-2x)(2+x^2) - (2-x^2)(4x)}{(2+x^2)^3} \\ &= \frac{2x(x^2-6)}{(2+x^2)^3}\end{aligned}$$

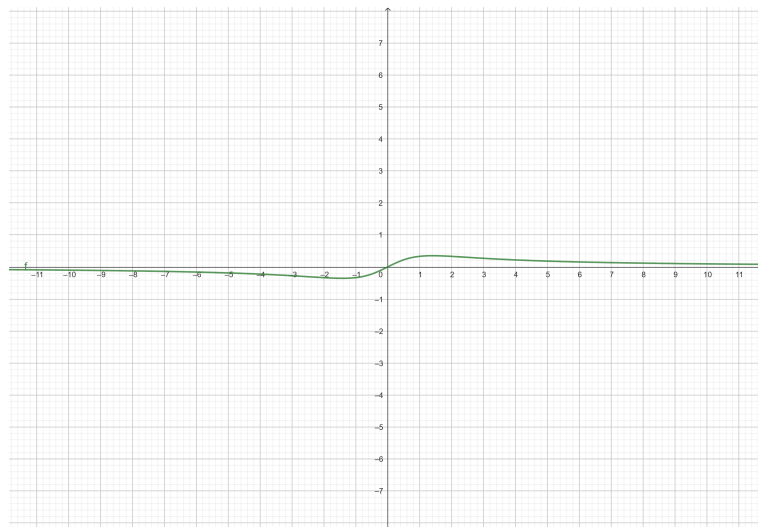
We have

$$\left. \frac{d^2y}{dx^2} \right|_{x=\sqrt{2}} < 0 \quad \text{and} \quad \left. \frac{d^2y}{dx^2} \right|_{x=-\sqrt{2}} > 0$$

Second derivative test shows:

y attains its maximum when $x = \sqrt{2}$.

Graphically, we have



Thus

$$\begin{cases} \tan \beta = \sqrt{2} \\ \tan \alpha = \frac{\sqrt{2}}{2+2} = \frac{\sqrt{2}}{4} \end{cases}$$

Hence, we have

$$\begin{cases} \beta = \arctan(\sqrt{2}) = 54.73561031^\circ \\ \alpha = \arctan\left(\frac{\sqrt{2}}{4}\right) = 19.47122063^\circ \end{cases}$$

Final Answer

$$\boxed{\alpha_{\max} = 19.47^\circ} \quad \boxed{\beta = 54.74^\circ}$$